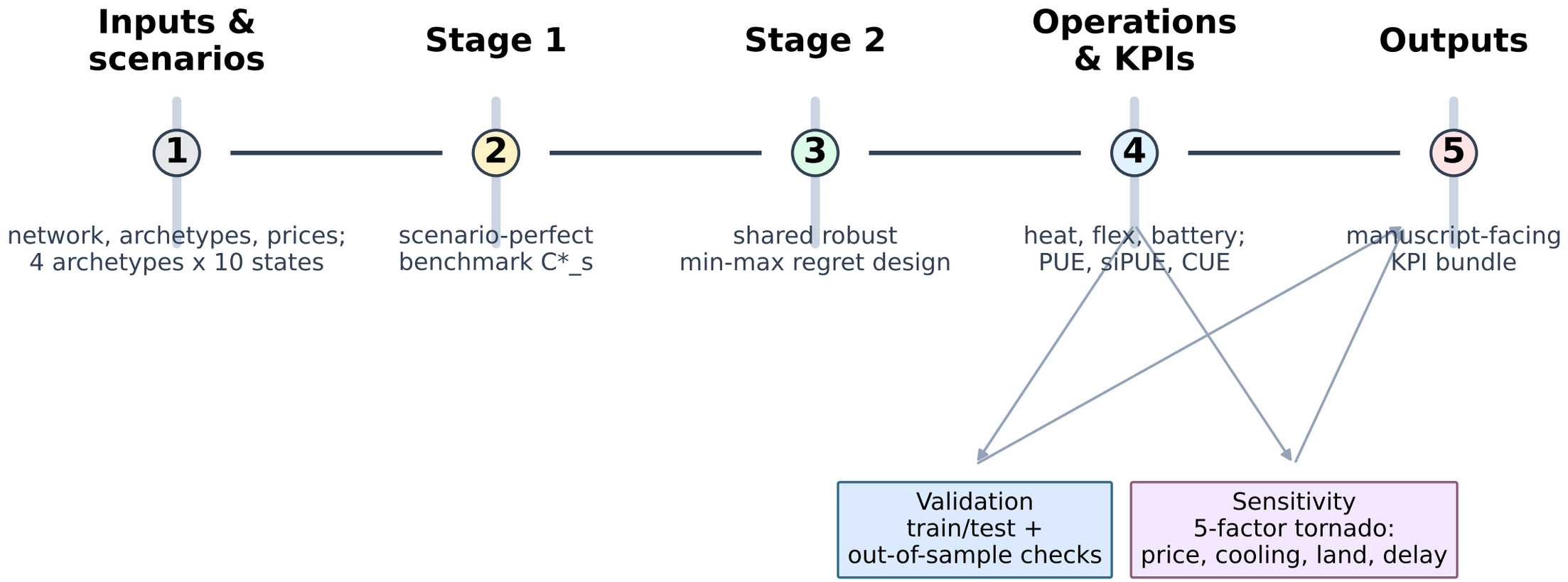
Full study roadmap: optimization, validation, and sensitivity together



The manuscript pipeline is not just the two-stage optimization: held-out validation and a five-factor sensitivity tornado both feed back into the reported KPI bundle.